BLABLA

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Abstract.

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1. Introduction

1.1. **Lattices.** For an introduction to lattices see [7, 10].

Definition 1. A lattice $\mathcal{L}$ is a discrete subgroup of $\mathbb{R}^m$.

So a lattice is a subgroup of the Abelian group $\langle \mathbb{R}^m, + \rangle$. Thus $\mathbf{0} \in \mathcal{L}$ and is discrete as a subset of $\mathbb{R}^m$. That is $\mathbf{0}$ is not a limit point. This means that the intersection of $\mathcal{L}$ and any bounded set of $\mathbb{R}^m$ is a finite set of points. For instance $\{\mathbf{0}\}$ and every subgroup of the additive group $\langle \mathbb{Z}^m, + \rangle$ are lattices of $\mathbb{R}^m$. A set of vectors $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\} \subset \mathbb{R}^m$ (where $n \leq m$), is called generator of $\mathcal{L}$ if each vector of $\mathcal{L}$, is a linear combination of vectors in $\mathcal{B}$ with integer coefficients. If the vectors of this set are also independent, we call $\mathcal{B}$ a basis of the lattice. In this case the number n is the same for all bases and we call it *rank* of the lattice, i.e. $n = \dim(span(\{\mathbf{b}_1, \ldots, \mathbf{b}_n\}))$. The integer m is called *dimension* of the lattice. If $n = m$, then we call the lattice, *full rank lattice*. Some authors call the rank as the dimension of the lattice. Usually we consider lattices which contain vectors only with integer coordinates. Such lattices are called *integer lattices*. A basis $\mathcal{B}$ is given by an $n \times m$ matrix, with rows the vectors of a basis of a lattice. If we have two distinct basis-matrices B and B', then there is a unimodular integer (square) matrix U with dimension n, such that $B = UB'$. It turns out that the Gram determinant $\det(BB^T)$ does not depend on the basis B or B'. This number is always positive and its positive square root is called determinant or volume of the lattice $\mathcal{L}$.

Definition 2. We call determinant or volume of the lattice $\mathcal{L}$ the positive real number

$$\det(\mathcal{L}) = vol(\mathcal{L}) = \sqrt{\det(BB^T)}.$$

The volume is in fact the (geometric) volume of the parallelepiped spanned by any basis of the lattice.

If $\mathcal{L}$ is a full rank lattice, then $vol(\mathcal{L}) = \det(B)$, where B is the square matrix representing a basis of $\mathcal{L}$. Having a matrix B we can write the lattice generated by the basis $\mathcal{B}$ as:

$$\mathcal{L}(\mathcal{B}) = \Big\{ \sum_{j=1}^{n} k_j \mathbf{b}_j : k_j \in \mathbb{Z} \Big\} \text{ or } \{\mathbf{x}B : \mathbf{x} \in \mathbb{Z}^n\}.$$

Since a lattice is a discrete subset of $\mathbb{R}^m$, then there is a nonzero vector $\mathbf{x} \in \mathcal{L}$ with the lowest Euclidean norm. We denote this norm by $\lambda_1(\mathcal{L})$ and we call it *first successive minima*. In general, not only one vector $\mathbf{v} \in \mathcal{L} - \{\mathbf{0}\}$ has $||\mathbf{v}|| = \lambda_1$. The number $|\{\mathbf{v} \in \mathcal{L} - \{\mathbf{0}\} : ||\mathbf{v}|| = \lambda_1\}|$ is called *kissing number* of the lattice. This number is at least 2 since, also $-\mathbf{v}$ has length λ_1. Further, Hermite proved that $\lambda_1(\mathcal{L}) \leq \sqrt{n} vol(\mathcal{L})^{1/n}$.

We set $\overline{B}_m(r) = \{\mathbf{x} \in \mathbb{R}^m : ||\mathbf{x}|| \leq r\}$. Then, $\lambda_1(\mathcal{L})$ is the smallest real number r such that, the real vector space $M_r(\mathcal{L}) = span(\mathcal{L} \cap \overline{B}_m(r))$, has at least one independent vector. Equivalently $\dim(M_r(\mathcal{L})) \geq 1$. In general we define $\lambda_j(\mathcal{L})$ to be the positive number $\inf\{r : \dim(M_r(\mathcal{L})) \geq j\}$. Always we have

$$\lambda_1(\mathcal{L}) \leq \lambda_2(\mathcal{L}) \leq \cdots \leq \lambda_n(\mathcal{L}).$$

An upper bound for the λ_i's was given by Minkowski.

Proposition 1.1. *(Second Theorem of Minkowski) For every integer $n > 1$ there exist a positive constant γ_n such that for any lattice with rank n and every $j \in \{1, 2, .., n\}$, we have the following inequality*

$$\Big(\prod_{i=1}^{n} \lambda_i\Big)^{1/n} \leq \sqrt{\gamma_n} vol(\mathcal{L})^{1/n}.$$

Also, the following Proposition holds.

Proposition 1.2. *Let $\mathcal{L}$ be a lattice of rank n with successive minima $\lambda_1, \ldots, \lambda_n$. Then there exist linearly independent vectors $\mathbf{b}_1, \ldots, \mathbf{b}_n$ such that $||\mathbf{b}_i|| = \lambda_i$ for $i = 1, \ldots, n$.*

Proof. See [10, Theorem 1.2]. □

The problem of finding a shortest vector in a lattice is called Shortest Vector Problem (SVP) and is one of the most important and difficult problems in lattices. Since this problem is hard (is proved that is NP-hard under randomized reductions by Ajtai [1]), we try to attack the approximation problem. For instance we are looking for lattice vectors $\mathbf{v}$ such that $||\mathbf{v}|| < \gamma(n)\lambda_1(\mathcal{L})$, where $n = rank(\mathcal{L})$. In fact LLL algorithm solves SVP for approximating factors $O(2^n)$.

The *covering radius* $\rho(\mathcal{L})$ of a lattice $\mathcal{L}$, is the minimal radius r such that, the union of balls of radius r centered at the points of the $\mathcal{L}$, cover $\mathbb{R}^n$. For instance $\rho(\mathbb{Z}^2) = \sqrt{2}/2$. Also, the quantity $\lambda_1/2$ is called *packing radius*. For $\mathbb{Z}^2$ is $\sqrt{2}/4$.

Also, the dual lattice $\mathcal{L}^*$ of $\mathcal{L}$ is defined,

$$\mathcal{L}^* = \{\mathbf{y} \in span(\mathcal{L}) : \mathbf{x} \cdot \mathbf{y} \in \mathbb{Z}, \text{ for every } \mathbf{x} \in \mathcal{L}\}.$$

For the determinant we have $\det(\mathcal{L}^*) \cdot \det(\mathcal{L}) = 1$.

1.2. **Voronoi cell of a lattice.** Let $\mathcal{L}$ be a lattice of $\mathbb{R}^n$. We provide some basic definitions.

Definition 3. The set

$$V(\mathcal{L}) = \{\mathbf{x} \in \mathbb{R}^n : \text{for every } \mathbf{y} \in \mathcal{L}, ||\mathbf{x}|| \leq ||\mathbf{x} - \mathbf{y}||\}$$

is called (closed) Voronoi cell of the lattice $\mathcal{L}$. That is, Voronoi cell of $\mathcal{L}$ is the set of all points in $\mathbb{R}^n$ that are at least as close to $\mathbf{0}$ as to any other lattice vector. On the other hand the set

$$V'(\mathcal{L}) = \{\mathbf{x} \in \mathbb{R}^n : \text{for every } \mathbf{y} \in \mathcal{L} - \{\mathbf{0}\}, ||\mathbf{x}|| < ||\mathbf{x} - \mathbf{y}||\}$$

is called open Voronoi cell of the lattice L. Both $V(\mathcal{L})$ and $V'(\mathcal{L})$ are non empty sets since $\mathbf{0}$ is in both sets. The Voronoi cell of a point lattice $\mathbf{v}$ is defined as $\mathbf{v}+V(\mathcal{L})$.

The following Lemma describes additional properties of $V(\mathcal{L})$.

Lemma 1.3. (**i**). *The (closed/open) Voronoi cell of a lattice is symmetric.*
(**ii**). $\overline{B}_n(\lambda_1/2) \subset V'(\mathcal{L}) \subset \overline{B}_n(\rho(\mathcal{L}))$, *where* $\rho(\mathcal{L})$ *is the covering radius of the lattice* $\mathcal{L}$. *So,* $V(\mathcal{L})$ *is bounded.*
(**iii**). $vol(\mathcal{L}) = vol(V(\mathcal{L})) = vol(V'(\mathcal{L}))$.

Proof. [10, Proposition 8.4]. □

From (**ii**) we conclude that if $\lambda_1 \approx 2\rho(\mathcal{L})$ then the Voronoi cell of $\mathcal{L}$ is close to a sphere.

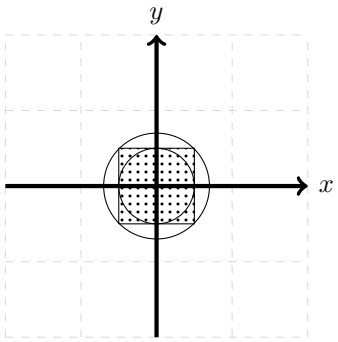


FIGURE 1. The rectangular is the voronoi cell of $\mathbb{Z}^2$. The inner circle is the border of the ball $\overline{B}_2(\lambda_1(\mathbb{Z}^2)/2)$ and the exterior circle is the border of the ball $\overline{B}_2(\rho(\mathbb{Z}^2))$, $\rho(\mathbb{Z}^2) = \sqrt{2}/2$.

Lemma 1.4. *Let* $\mathbf{v} \in \mathcal{L} - \{\mathbf{0}\}$. *We define* $H(\mathbf{v}) = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x}\cdot\mathbf{v} \leq \frac{1}{2}\mathbf{v}\cdot\mathbf{v}\}$. *Then,*

$$V(\mathcal{L}) = \bigcap_{\mathbf{v}\in\mathcal{L}-\{\mathbf{0}\}} H(\mathbf{v}).$$

Proof. $\mathbf{x} \in V(\mathcal{L}) \Leftrightarrow ||\mathbf{x}||^2 \leq ||\mathbf{x}-\mathbf{v}||^2 \Leftrightarrow \mathbf{x}\cdot\mathbf{x} \leq (\mathbf{x}-\mathbf{v})\cdot(\mathbf{x}-\mathbf{v}) \Leftrightarrow \mathbf{x}\cdot\mathbf{v} \leq \frac{1}{2}\mathbf{v}\cdot\mathbf{v}$. Since the previous equivalences are valid for every $\mathbf{v} \in L$ the Lemma follows. □

From the proof we see that

$$H(\mathbf{v}) = \{\mathbf{x} \in \mathbb{R}^n : ||\mathbf{x}|| \leq ||\mathbf{x}-\mathbf{v}||\} \ (\mathbf{v} \in \mathcal{L} - \{\mathbf{0}\}).$$

Also, a simple consequence of Lemma 1.4 is that the Voronoi cell of a lattice is a convex subset of $\mathbb{R}^n$. A minimal set of lattice vectors, say $RV(\mathcal{L}) \subset \mathcal{L}$, also generates $V(\mathcal{L})$. The vectors in $RV(\mathcal{L})$ are called *relevant vectors*. A similar property holds for open Voronoi cells of a lattice. That is,

$$V'(\mathcal{L}) = \bigcap_{\mathbf{v}\in\mathcal{L}-\{\mathbf{0}\}} H'(\mathbf{v}),$$

where

$$H'(\mathbf{v}) = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x}\cdot\mathbf{v} < \frac{1}{2}\mathbf{v}\cdot\mathbf{v}\}.$$

Now we set $H_{\mathbf{v}} = H(\mathbf{v}) - H'(\mathbf{v})$. That is,

$$H_{\mathbf{v}} = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \cdot \mathbf{v} = \frac{1}{2}\mathbf{v} \cdot \mathbf{v}\} = \{\mathbf{x} \in \mathbb{R}^n : ||\mathbf{x}|| = ||\mathbf{x} - \mathbf{v}||\}.$$

This set defines a hyperplane of $\mathbb{R}^n$. In fact, $H_{\mathbf{v}}$ is the bisector of the line segment $\mathbf{0}, \mathbf{v}$ in $\mathbb{R}^n$. We say that a lattice vector $\mathbf{v}$ is a Voronoi vector if and only if $H_{\mathbf{v}} \cap V(\mathcal{L}) \neq \emptyset$. To find the Voronoi vectors, suffices to find the solutions of the equation (with $2n$ unknowns),

$$\sum_{i=1}^{n} v_i^2 = 2\sum_{i=1}^{n} x_i v_i, \tag{1.1}$$

under the constraints $\mathbf{x} = (x_i)_i \in \mathbb{R}^n$ and $\mathbf{v} = (v_i)_i \in \mathcal{L}$. The unknowns are x_i, v_i. Another definition of *relevant vectors* is the following. A *Voronoi vector* is called *relevant* vector if and only if the intersection $H_{\mathbf{v}} \cap V(\mathcal{L})$ is a $(n-1)-$dimensional face of $V(\mathcal{L})$ (also called *facet* of $V(\mathcal{L})$). To find the Voronoi Relevant vectors we need exponential time. We have the following.

Proposition 1.5. *There is an algorithm which on input a basis of $\mathcal{L}$, outputs the Voronoi relevant vectors in time $O(2^{3.5n})$ and space $O(2^n)$.*

Proof. [9, Proposition 3.9] □

Example 1.6. Consider the lattice $\mathcal{L} = \mathbb{Z}^2$. The vector $\mathbf{v} = \mathbf{e}_1 + \mathbf{e}_2 = (1,1)$ is a Voronoi vector. Indeed, from formula (1.1) we get $2 = 2(x_1+x_2)$ so $x_1+x_2 = 1$. This line (which is $H_{\mathbf{v}}$) intersects the Voronoi cell of $\mathbb{Z}^2$ to the point $(1/2, 1/2)$. So $\mathbf{v} = (1,1)$ satisfies the definition. We used that $V(\mathbb{Z}^2) = [-1/2, 1/2] \times [-1/2, 1/2]$. We can prove that the following eight vectors $(\pm 1, 0), (0, \pm 1), (\pm 1, \pm 1)$ are all Voronoi vectors. We can see that the vectors $(\pm 1, 0)$ and $(0, \pm 1)$ are relevant. We show this for $(1,0)$. Clearly $H_{(1,0)} = \{(x_1, x_2) : x_1 = 1/2\}$. So, $H_{(1,0)} \cap V(\mathbb{Z}^2) = [-1/2, 1/2]$. That is, the intersection is one dimensional face of $V(\mathbb{Z}^2)$. All the vectors $\pm\mathbf{e}_i$ are relevant (also called *strict*). The rest vectors which are not relevant are called *lax*. From the definition of relevant vectors we get that the relevant vectors are 1-1 with the facets of $V(\mathcal{L})$. So for $\mathbb{Z}^2$ we have exactly 4 relevant vectors, $\pm\mathbf{e}_1$ and $\pm\mathbf{e}_2$.

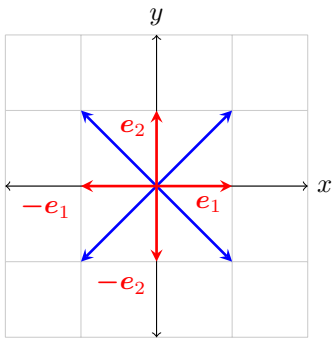


FIGURE 2. These 8 vectors are all the Voronoi vectors of $\mathbb{Z}^2$. The red vectors $(\pm\mathbf{e}_i)$ are the relevant vectors of $\mathbb{Z}^2$.

We have the following lemma concerning relevant vectors.

Lemma 1.7. $\mathbf{v} \in \mathcal{L} - \{\mathbf{0}\}$ *is relevant if and only if*

$$\mathbf{x} \cdot \mathbf{v} < \mathbf{x} \cdot \mathbf{x},$$

for every $\mathbf{x} \in \mathcal{L} - \{\mathbf{0}, \mathbf{v}\}$.

Proof. In [12] Voronoi showed that $\mathbf{v}$ is relevant if and only if $\mathbf{0}$ and $\mathbf{v}$ are the only closest vectors to $\frac{1}{2}\mathbf{v}$. I.e. $\mathbf{v}$ is relevant if and only if $CVP(\mathcal{L}, \frac{1}{2}\mathbf{v}) = \{\mathbf{0}, \mathbf{v}\}$. Equivalently, $\mathbf{v}$ is relevant if and only if $d(\mathbf{x}, \frac{1}{2}\mathbf{v}) > d(\mathbf{v}, \frac{1}{2}\mathbf{v})$ for every $\mathbf{x} \in \mathcal{L} - \{\mathbf{0}, \mathbf{v}\}$.

Observe that

$$||\mathbf{x}||^2 - \mathbf{v} \cdot \mathbf{x} = ||\frac{1}{2}\mathbf{v} - \mathbf{x}||^2 - ||\frac{1}{2}\mathbf{v}||^2.$$

Thus, $\mathbf{x} \cdot \mathbf{v} < \mathbf{x} \cdot \mathbf{x}$, if and only if $||\frac{1}{2}\mathbf{v} - \mathbf{x}|| > ||\frac{1}{2}\mathbf{v}|| = d(\mathbf{0}, \frac{1}{2}\mathbf{v}) = d(\mathbf{v}, \frac{1}{2}\mathbf{v})$. The Lemma follows. □

Example 1.8. If $\mathcal{L} = \mathbb{Z}^2$ we found in example 1.6 that $\pm\mathbf{e}_i$ $(i = 1, 2)$ are relevant. Indeed, using the previous Lemma, we get that $\mathbf{x} \cdot (\pm\mathbf{e}_i) = \pm x_i < ||\mathbf{x}||^2 = x_1^2 + x_2^2$ for every $\mathbf{x} = (x_1, x_2) \in \mathbb{Z}^2 - \{(0, 0), \pm\mathbf{e}_i\}$.

Voronoi also proved the following.

Proposition 1.9. (**i**). $\mathbf{v}$ *is a Voronoi vector if and only if* $\mathbf{v}$ *is the only shortest vector in the coset* $2\mathcal{L} + \mathbf{v}$.
(**ii**). $\mathbf{v}$ *is relevant-Voronoi vector if and only if* $\pm\mathbf{v}$ *are the only shortest vectors in the coset* $2\mathcal{L} + \mathbf{v}$.

Proof. [5, Theorem 2] □

1.2.1. *Characters and discrete Fourier over* $\mathcal{L}/2\mathcal{L}$. We provide some useful definitions.

Definition 4. We call *vonorm* (Voronoi norm) of a coset of $\mathbf{x} + 2\mathcal{L}$ the square Euclidean length of a shortest vector in the coset. We denote this by $\mathcal{V}(\bar{\mathbf{x}})$. I.e.

$$\mathcal{V} : \mathcal{L}/2\mathcal{L} \to \mathbb{Z}_{\geq 0},$$

$$\mathcal{V}(\bar{\mathbf{x}}) = \min\{||\mathbf{y}||^2 : \mathbf{y} \in \mathbf{x} + 2\mathcal{L}\}.$$

The vonorm of the trivial coset $\mathbf{0} + 2\mathcal{L}$ is zero and we call this vonorm improper. The other vonorms are called proper[1].

Definition 5. We call *vonorm* space of $\mathcal{L}$, the $\mathbb{F}_2$ vector space $\mathrm{Vo}(\mathcal{L}) = \mathcal{L}/2\mathcal{L}$.

Definition 6. A map $\chi : \mathcal{L} \to \{\pm 1\}$ is called a character over $\mathcal{L}$ if and only if $\chi(\mathbf{v} + \mathbf{w}) = \chi(\mathbf{v}) \cdot \chi(\mathbf{w})$ (i.e. is a group homomorphism from the additive group $(\mathcal{L}, +)$ to the multiplicative group $(\{\pm 1\}, \cdot)$).

Here with character we mean a *real* character. In general a character over a finite Abelian group G, is an homomorphism $\chi : G \to S^1$ where $S^1 = \{z \in \mathbb{C} : |z| = 1\}$. If G is additive, then $\chi(g_1 + g_2) = \chi(g_1)\chi(g_2)$ and $\chi(0) = 1$. Also, $\chi(-g) = (\chi(g))^{-1}$. The constant map $\chi(g) = 1$ for evey g in G is a character. Is called trivial character. Here we consider only real characters $\chi : G \to \{\pm 1\}$. Also, the dual group of G (also is called character group of G) is the set of all homomorphishm $G \to \{\pm 1\}$.

[1] in [6] the word norm means the square Euclidean length

We denote the dual group $\hat{G}$. This is indeed a group using the following operation, if $\chi, \psi \in \hat{G}$ then $(\chi\psi)(g) = \chi(g)\psi(g)$ for every $g \in G$. So, the set

$$\{\chi : \chi \text{ is a (real) character of } \mathcal{L}\}$$

is a group which we denote it by $\widehat{\mathcal{L}/2\mathcal{L}}$. Also, it's a $\mathbb{R}-$vector space which is isomorphic with the dual real vector space of $\mathcal{L}/2\mathcal{L}$. That is, the set of characters of $\mathcal{L}$ it's the dual of the vonorm space. Also,

Lemma 1.10. *If χ is non trivial, then*

$$\sum_{g \in \mathrm{Vo}(\mathcal{L})} \chi(g) = 0.$$

So the number of $+1$'s and -1's are the same as g runs in $\mathrm{Vo}(\mathcal{L}) = \mathcal{L}/2\mathcal{L}$.

Fourier Transforms on finite Abelian groups.
If we have a finite Abelian group G (you can think G to be $\widehat{\mathcal{L}/2\mathcal{L}}$) and a real valued function $f : G \to \mathbb{R}$ then the discrete Fourier of f is defined as the sum

$$\widehat{f}(\chi) = \sum_{g \in G} f(g)\chi(g)$$

and the inverse is

$$f(y) = \sum_{\chi \in \hat{G}} \widehat{f}(\chi)\chi(y).$$

as you wrote it, orthogonal are also obtuse superbases

1.3. **Voronoi First Kind (VFK) Lattices.**

Definition 7. A lattice $\mathcal{L}$ of dimension n is called Voronoi First Kind (VFK) if and only if there is a basis $\{\mathbf{v}_j\}_{j=1,\ldots,n}$ and a lattice vector $\mathbf{v}_0$ such that :
(**i**). $\sum_{j=0}^{n} \mathbf{v}_j = \mathbf{0}$.
(**ii**). $\mathbf{v}_i \cdot \mathbf{v}_j \leq 0$ for $i, j = 0, 1, ..., n$ with $i \neq j$.

The values of the inner products $p_{ij} = -\mathbf{v}_i \cdot \mathbf{v}_j$ are called *Selling* parameters[2]. A basis that satisfies the previous definition is called *obtuse superbase*. Also, the first condition of the definition is called *superbase condition* and the second *obtuse condition*. If $p_{ij} = -\mathbf{v}_i \cdot \mathbf{v}_j > 0$, for all i, j $(i \neq j)$, then we say that we have a *strict obtuse superbase*.
Having the $n(n+1)/2$ quantities $p_{ij}, 0 \leq i < j \leq n$, are enough to determine the structure of the lattice. Indeed, this is immediate if we use the formula,

$$||\mathbf{v}_i||^2 = \sum_{k=0,(k\neq i)}^{n} p_{ik}. \tag{1.2}$$

For instance for $n = 2$ and $i = 0$ we get

$$p_{01} + p_{02} = -\mathbf{v}_0 \cdot \mathbf{v}_1 - \mathbf{v}_0 \cdot \mathbf{v}_2 = (\mathbf{v}_1 + \mathbf{v}_2) \cdot \mathbf{v}_1 + (\mathbf{v}_1 + \mathbf{v}_2) \cdot \mathbf{v}_2 =$$
$$(\mathbf{v}_1 + \mathbf{v}_2)^2 = ||\mathbf{v}_0||^2.$$

Also we have the Selling formula,

$$||x_0\mathbf{v}_0 + \cdots + x_n\mathbf{v}_n||^2 = \sum_{0 \leq i < j \leq n} p_{ij}(x_i - x_j)^2. \tag{1.3}$$

[2]There is a minor typo in formula (5) of [5], p_{ij} (which are defined in formula (1) of the same paper) are the minus dot products of the basis vectors and not $\mathbf{v}_i \cdot \mathbf{v}_j$

So, for instance if $n=3$, then $||\mathbf{v}_0+\mathbf{v}_1||^2 = p_{02}+p_{03}+p_{12}+p_{13}$. We write the last sum as $p_{01|23}$. Besides, we use the symbol $p_{i|jkl} = p_{ij}+p_{ik}+p_{il}$. Also, $p_{i|jkl} = p_{jkl|i}$ because $p_{ab} = p_{ba}$. So we have defined the following symbols,

$$p_{ij|kl} = p_{ik}+p_{il}+p_{jk}+p_{jl} \tag{1.4}$$

$$p_{i|jkl} = p_{ij}+p_{ik}+p_{il} \tag{1.5}$$

for distinct i,j,k,l in the set $\{0,1,2,3\}$. Furthermore,

$$p_{i|jkl} = p_{i|\sigma_3(j,k,l)},\ p_{ij|kl} = p_{ij|\sigma_2(k,l)} \tag{1.6}$$

for any permutations σ_3(or σ_2) of three (two symbols resp.). Some simple properties of the previous symbols are the following.

Lemma 1.11. *Let $\mathcal{L}$ be a VFK (full rank) lattice of dimension 3 and $\{\mathbf{v}_i\}_{i=0,1,2,3}$ be an ~~obtuse~~ superbasis. For distinct $i,j,k,\ell \in \{0,1,2,3\}$ we get,*

(**i**).
$$p_{ii}+p_{ij}+p_{ik}+p_{i\ell} = 0.$$

(**ii**).
$$||\mathbf{v}_i+\mathbf{v}_j||^2 = p_{ij|k\ell}.$$

(**iii**).
$$||\mathbf{v}_i+\mathbf{v}_j+\mathbf{v}_k||^2 = p_{\ell|ijk}.$$

(**iv**).
$$||\mathbf{v}_i||^2 = p_{\ell i}+p_{ki}+p_{ji} = p_{i|jk\ell}.$$

(**v**).
$$||\alpha\mathbf{v}_i+\beta\mathbf{v}_j||^2 = \alpha^2 p_{i|k\ell}+\beta^2 p_{j|k\ell}.$$

Proof. (**i**). Say $i=0$. Then,

$$p_{00}+p_{01}+p_{02}+p_{03} = \mathbf{v}_0\cdot(\mathbf{v}_0+\mathbf{v}_1+\mathbf{v}_2+\mathbf{v}_3) = \mathbf{v}_0\cdot\mathbf{0} = 0.$$

(**ii**), (**iii**), (**iv**), and (**v**), are consequences of Selling formula. □

Remark 1.12. The first part of the previous lemma is written

$$p_{ii}+p_{i|jkl} = 0,$$

for distinct $i,j,k,l \in \{0,1,2,3\}$.

Using Selling formula we can compute the norms of all the vectors in the finite group $\mathcal{L}/2\mathcal{L}$. Then, using Proposition 1.9 we can find all the Voronoi vectors of a VFK lattice. So, if we substitute in Selling formula the x_i's with even integers we just choose different representative of $\mathbf{v}$ in $2\mathcal{L}$. Thus, to compute the norms of $\mathcal{L}/2\mathcal{L}$ we substitute to 0 whenever x_i is even and 1 whenever x_i is odd. In the case we have an obtuse superbase, the norm of the vectors of $\mathcal{L}/2\mathcal{L}$ are the elements of the set

$$\mathcal{A}_n = \{p_{0|123\ldots n}, p_{1|02\ldots n}, p_{2|013\ldots n}, \ldots, p_{12|03\ldots n}, p_{13|02\ldots n}, \ldots\}$$

and so the vectors of $\mathcal{L}/2\mathcal{L}$ are,

$$\mathbf{v}_k, \text{with } k\in\{0,1,\ldots,n\}, \text{ and}$$
$$(\mathbf{v}_i+\mathbf{v}_j), 1\le i<j\le n$$

and so on. I.e. all the vectors

$$\sum_{i\in I\neq\emptyset,\ I\subset\{0,1,\ldots,n\}} \mathbf{v}_i.$$

So in the case we have a VFK lattice we can compute $\lambda_1(\mathcal{L})$ by computing all the elements of the set $\mathcal{A}_n$. Since

$$|\mathcal{A}_n| = \binom{n+1}{1} + \binom{n+1}{2} + \cdots + \binom{n+1}{n} = 2^{n+1} - 2,$$

this provide us with exponential time. Remark that this naive brute force is better than the best SVP algorithm which has running time $O(2^{n+o(n)})$ [2]. In fact in [] they managed to get a polynomial time algorithm.

1.4. **Characters.** We have the following Theorem.

Theorem 1.13. *The characters of a VFK lattice $\mathcal{L}$ correspond to the subsets S of $\{0,1,2,...,n\}$ such that $|S|$ is even. Furthermore they are defined by,*

$$\chi_S(\mathbf{v}_i) = \begin{cases} -1, & i \in S, \\ 1, & i \notin S. \end{cases}$$

Proof. [5, Theorem 5]. Since χ_S is a character and the number of subsets S of $\{0,1,...,n\}$ with $|S|$ even, is $\frac{1}{2}2^{n+1} = 2^n$, these are all the characters of lattice $\mathcal{L}$. □

We define the conjugate norm $\mathcal{C}$ of a character to be,

$$\mathcal{C} : \widehat{\mathcal{L}/2\mathcal{L}} \to \mathbb{Z}_{\geq 0},\ \mathcal{C}(\chi) = -\frac{1}{2^{n-1}} \sum_{\bar{\mathbf{v}} \in \mathcal{L}/2\mathcal{L}} \chi(\bar{\mathbf{v}})\mathcal{V}(\bar{\mathbf{v}}) \tag{1.7}$$

where $\mathcal{V}$ is the vonorm map (see definition 4). This map is called conorm and it is the discrete fourier of $\mathcal{V}$ (up to a scaling factor). Note that we did not assume in the previous definition that our lattice is VFK. If we consider the inversion fourier formula we get

$$\mathcal{V}(\bar{\mathbf{v}}) = \sum_{\chi : \chi(\bar{\mathbf{v}}) = -1} \mathcal{C}(\chi)$$

We have the following lemma for VFK lattices.

Lemma 1.14. *If $\mathcal{L}$ is a VFK lattice, the proper conorms are given by*

$$\mathcal{C}(\chi_S) = \begin{cases} p_{ij}, & S = \{i,j\}, \\ 0, \textit{otherwise}. \end{cases}$$

todo:do the proof

Proof. [5, Theorem 6]. We can see an easy case for $n = 2$ (rank 2 lattices are VFK, see subsection 1.5). Say that we want to compute $\mathcal{C}(\chi_{\{0,1\}})$. We first note that the Voronoi vectors of $\mathcal{L}$ are,

$$\{\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_0 + \mathbf{v}_1, \mathbf{v}_0 + \mathbf{v}_2, \mathbf{v}_1 + \mathbf{v}_2\}.$$

So, the representatives of $\mathcal{L}/2\mathcal{L}$ are,

$$\{\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2\}.$$

Then,

$$\mathcal{C}(\chi_{\{0,1\}}) = -\frac{1}{2}\big(\chi_{\{0,1\}}(\mathbf{v}_0)||\mathbf{v}_0||^2 + \chi_{\{0,1\}}(\mathbf{v}_1)||\mathbf{v}_1||^2 + \chi_{\{0,1\}}(\mathbf{v}_2)||\mathbf{v}_2||^2\big) =$$

$$-\frac{1}{2}\big(-||\mathbf{v}_0||^2 - ||\mathbf{v}_1||^2 + ||\mathbf{v}_2||^2\big) =$$

$$\frac{1}{2}\big(||\mathbf{v}_0||^2 + ||\mathbf{v}_1||^2 - ||\mathbf{v}_2||^2\big) =$$

$$\frac{1}{2}\big(||\mathbf{v}_1+\mathbf{v}_2||^2+||\mathbf{v}_1||^2-||\mathbf{v}_2||^2\big)=$$

$$\frac{1}{2}\big(2||\mathbf{v}_1||^2+2\mathbf{v}_1\cdot\mathbf{v}_2\big)=\mathbf{v}_1\cdot(\mathbf{v}_1+\mathbf{v}_2)=-\mathbf{v_1}\cdot\mathbf{v}_0=p_{01}$$

□

This Lemma explains the scaling factor in the formula (1.7) of $\mathcal{C}(\chi)$. Also, the *conorm space* consists from the Selling parameters for $0\leq i<j\leq n$. If we do not have an obtuse superbase in a VFK lattice, we can start from any superbase $\{\mathbf{v}_0,...,\mathbf{v}_n\}$ and *transfer* all the computations to the conorm space. Then, applying simple computations to p_{ij} we can make them all positive, and go back to the vonorm space to compute the new superbase, which finally will be obtuse.

Examples
(**i**). The lattice $\mathbb{Z}^n$ is a VFK lattice. The volume of $\mathbb{Z}^n$ is 1 and Voronoi cells are the hypercubes with vertices $(\pm1/2,\pm1/2,...,\pm1/2)$, and the relevant vectors are the $2n$ vectors $\pm\mathbf{e}_i$. The covering radius is $\rho(\mathbb{Z}^n)=\frac{\sqrt{n}}{2}$ and the first successive minima $\lambda_1=1$.

(**ii**). The root lattice $A_n\subset\mathbb{Z}^{n+1}$ generated by the n vectors

$$\{\mathbf{v}_1=(1,-1,...,0),\mathbf{v}_2=(0,1,-1,0,...,),...,\mathbf{v}_n=(0,0,...,1,-1)\}\subset\mathbb{Z}^{n+1}.$$

is a VKF lattice, since the previous basis provide an obtuse superbasis. Indeed,

$$\mathbf{v}_0=-\mathbf{v}_1-\mathbf{v}_2-\cdots-\mathbf{v}_n=(-1,...,1).$$

It is easy to check $\mathbf{v}_i\cdot\mathbf{v}_j\leq0$, $i\neq j$. Another description of this lattice is,

$$A_n=\{(x_0,x_1,...,x_n)\in\mathbb{Z}^{n+1}:\sum_{i=0}^{n}x_i=0\}.$$

The volume of this lattice is $vol(A_n)=\sqrt{n+1}$. The covering radius is $\rho(A_n)=\frac{\alpha(n+1-\alpha)}{n+1}$ where α the integer part of $(n+1)/2$. Also, $\lambda_1=\sqrt{2}$.

(**iii**). The lattice $D_3\subset\mathbb{Z}^3$ with generating set the three vectors

$$\{(1,-1,0),(0,1,-1),(0,1,1)\}\subset\mathbb{Z}^3,$$

is again a VFK lattice. The relevant vectors are $\mathbf{e}_i\pm\mathbf{e}_j$. In fact D_3 is isomorphic with A_3. For $n\geq4$ the lattices D_n are not VFK [3, Theorem B.2.6, p.92]. Furthermore, the dual lattices D_n^* $(n\geq4)$, E_i^*, $i=6,7,8$ are not VFK. $E_8\subset\mathbb{R}^8$ is the following lattice,

$$\big\{\mathbf{x}\in\mathbb{Z}^8:\sum_{i=1}^{8}x_i\equiv0\bmod2\big\}.$$

Since E_8^* and E_8 are isomorphic, also E_8 is not VFK.

To check if a basis of a (full rank) lattice of dimension n is obtuse superbase, we need to check the sign of $\frac{n(n-1)}{2}+n$ dot products of the definition. So for large n we need $O(n^2)$ dot product operations.

1.5. **Dimension 2.** It is very easy to show that all the integer lattices of dimension 2 are VFK lattices (see [3, subsection B.2.1]). We begin with the following definition.

Definition 8. An ordered basis $\{\mathbf{b}_1, \mathbf{b}_2\} \subset \mathbb{Z}^2$ of a lattice $\mathcal{L}$ is called reduced if and only if $||\mathbf{b}_1||, ||\mathbf{b}_2|| \leq ||\mathbf{b}_1 \pm \mathbf{b}_2||$.

Proposition 1.15. *An ordered basis $\{\mathbf{b}_1, \mathbf{b}_2\} \subset \mathbb{Z}^2$ is reduced if and only if $||\mathbf{b}_i|| = \lambda_i(\mathcal{L})$ for $i = 1, 2$.*

Corollary 1.16. *For a two dimensional lattice $\mathcal{L}$ there is always a reduced basis.*

Proof. Consequence from the previous Proposition and Proposition 1.2. □

Lemma 1.17. *All two dimensional lattices $\mathcal{L} \subset \mathbb{Z}^2$ are VFK.*

Proof. It is enough to find an obtuse superbase of $\mathcal{L}$. Let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ be a reduced basis of $\mathcal{L}$ (by corollary 1.16). Without loss of generality we assume that $\mathbf{b}_1 \cdot \mathbf{b}_2 \leq 0$. If not we choose the new basis $\{\mathbf{b}_1, -\mathbf{b}_2\}$ (which is again reduced). We set $\mathbf{b}_3 = -\mathbf{b}_1 - \mathbf{b}_2$. Then the set $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$, has the superbase property. We set also, $||\mathbf{b}_i||^2 = \ell_i$ $(i = 1, 2)$. Since $\mathcal{B}$ is reduced we get $\ell_i \leq ||\mathbf{b}_1 + \mathbf{b}_2||^2$, thus $-\mathbf{b}_1 \cdot \mathbf{b}_2 \leq \frac{1}{2}\ell_i$. Then,

$$\mathbf{b}_i \cdot \mathbf{b}_3 = -\ell_i - \mathbf{b}_1 \cdot \mathbf{b}_2 \leq -\ell_i + \frac{\ell_i}{2} = -\frac{\ell_i}{2} < 0.$$

So the set $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$, has also the obtuse property. The Lemma follows. □

Note that there is an efficient algorithm, Gauss-Lagrange algorithm (polynomial bit complexity with respect the maximum norm of the input basis), that provides us with a reduced basis. So, there is an efficient algorithm to find an obtuse superbase basis for all integer lattices of dimension 2.

1.6. **Dimension 3.** The fact that all the rank three integer lattices are VFK is very old and was proved by Selling in 1874 [11] and improved by Voronoi in 1908 [12]. Furthermore, in [5] the authors provide a geometric proof. This provide us with an algorithm to find an obtuse superbasis. On the other hand, they did not provide any analysis or a specific pseudocode. To prove the existence of an obtuse superbasis of a rank 3 lattice is relatively easy. To provide an algorithm needs more work.

1.6.1. *Existence.* We follow [3, B.2.2, p.90]. Let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be a basis of $\mathcal{L}$ and $\{\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the corresponding superbasis. We set

$$\mathbf{v}_S = \sum_{a \in S} \mathbf{v}_a, \text{ for some set } S \subseteq \{0, 1, 2, 3\}$$

and $p_{ij} = -\mathbf{v}_i \cdot \mathbf{v}_j$. We assume that not all $p_{ij} = 0$. We have the following properties.

Lemma 1.18. (**i**).

$$\sum_{S \subset \{0,1,2,3\}} \mathbf{v}_S = \mathbf{0}.$$

(**ii**).

$$||\mathbf{v}_1||^2 + ||\mathbf{v}_2||^2 + ||\mathbf{v}_3||^2 - p_{12} - p_{13} - p_{23} = \sum_{0 \leq i < j \leq 3} p_{ij}.$$

(**iii**).

$$\sum_{S\subset\{0,1,2,3\},\ |S|=1} ||\mathbf{v}_S||^2 = 2(||\mathbf{v}_1||^2 + ||\mathbf{v}_2||^2 + ||\mathbf{v}_3||^2 - p_{12} - p_{13} - p_{23}).$$

(**iv**).

$$\sum_{S\subset\{0,1,2,3\},\ |S|=2} ||\mathbf{v}_S||^2 = 4(||\mathbf{v}_1||^2 + ||\mathbf{v}_2||^2 + ||\mathbf{v}_3||^2 - p_{12} - p_{13} - p_{23}).$$

(**v**).

$$\sum_{S\subset\{0,1,2,3\},\ |S|=3} ||\mathbf{v}_S||^2 = 2(||\mathbf{v}_1||^2 + ||\mathbf{v}_2||^2 + ||\mathbf{v}_3||^2 - p_{12} - p_{13} - p_{23}).$$

Proof. (**i**). It is straightforward.
(**ii**). We shall use relation (1.2). For instance

$$||\mathbf{v}_1||^2 = p_{10} + p_{12} + p_{13}.$$

Then we have,

$$||\mathbf{v}_1||^2 + ||\mathbf{v}_2||^2 + ||\mathbf{v}_3||^2 - p_{12} - p_{13} - p_{23} =$$
$$p_{10} + p_{12} + p_{13} + p_{20} + p_{21} + p_{23} + p_{10} + p_{12} + p_{13} -$$
$$p_{12} - p_{13} - p_{23} =$$
$$p_{10} + p_{12} + p_{13} + p_{20} + p_{23} + p_{30}.$$

The result follows.
We shall prove only (**iii**).
(**iii**).

$$\sum_{S\subset\{0,1,2,3\},\ |S|=1} ||\mathbf{v}_S||^2 = ||\mathbf{v}_0||^2 + ||\mathbf{v}_1||^2 + ||\mathbf{v}_2||^2 + ||\mathbf{v}_3||^2 =$$
$$||\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3||^2 + ||\mathbf{v}_1||^2 + ||\mathbf{v}_2||^2 + ||\mathbf{v}_3||^2 =$$
$$2||\mathbf{v}_1||^2 + 2||\mathbf{v}_2||^2 + 2||\mathbf{v}_3||^2 - 2p_{12} - 2p_{13} - 2p_{23}.$$

The result follows.
The cases (**iv**) and (**v**) are treated with similar way as (**iii**). □

For each superbasis $\{\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ of $\mathcal{L}$ we consider the quantity

$$0 < N(\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = \sum_{S\subset\{0,1,2,3\}} ||\mathbf{v}_S||^2 =$$

$$\sum_{S\subset\{0,1,2,3\},\ |S|=1} ||\mathbf{v}_S||^2 + \sum_{S\subset\{0,1,2,3\},\ |S|=2} ||\mathbf{v}_S||^2 + \sum_{S\subset\{0,1,2,3\},\ |S|=3} ||\mathbf{v}_S||^2.$$

From Lemma 1.18 (ii), (iii) and (iv), we get

$$N(\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = 8(||\mathbf{v}_1||^2 + ||\mathbf{v}_2||^2 + ||\mathbf{v}_3||^2 - p_{12} - p_{13} - p_{23})$$

and from (i) of the same lemma we finally get,

$$N(\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = 8 \sum_{0\le i<j\le 3} p_{ij}.$$

We remark that the set

$$\{N(\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) : \{\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \text{ superbasis}\} \subset \mathbb{N}.$$

Now, we claim that a superbasis that minimize $N(.)$ is an obtuse one. Indeed, say it's not. Assume that $(\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ minimizes N and is not obtuse, for instance

is it true that, multiplication with $SL_4(\mathbb{Z})$ sends a superbasis to a superbasis?

say that $p_{01} < 0$ (i.e. the angle between $\mathbf{v}_0$ and $\mathbf{v}_1$ is acute. Remark that we have defined $p_{ij} = -\mathbf{v}_i \cdot \mathbf{v}_j$). We set $B = [\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3]^T$. Pick the matrix $U_4 \in SL_4(\mathbb{Z})$ and consider $B' = U_4 B$, where

$$U_4 = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}.$$

Then, B' provides us with a new superbasis of $\mathcal{L}$.

$$U_4 B = B' = [\mathbf{v}'_0, \mathbf{v}'_1, \mathbf{v}'_2, \mathbf{v}'_3]^T = [-\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_0 + \mathbf{v}_2, \mathbf{v}_0 + \mathbf{v}_3]^T =$$
$$[-\mathbf{v}_0, \mathbf{v}_1, -\mathbf{v}_1 - \mathbf{v}_3, -\mathbf{v}_1 - \mathbf{v}_2]^T.$$

We note that $\sum_{j=0}^{3} \mathbf{v}'_k = \mathbf{0}$ (i.e. is a superbasis) and $\{\mathbf{v}'_1, \mathbf{v}'_2, \mathbf{v}'_3\}$ is a basis of $\mathcal{L}$. Furthermore,

$$p'_{01} = -\mathbf{v}'_0 \cdot \mathbf{v}'_1 = \mathbf{v}_0 \cdot \mathbf{v}_1 = -p_{01},$$
$$p'_{02} = -p_{00} - p_{02},\ p'_{03} = -p_{00} - p_{03},\ p'_{23} = p_{00} + p_{03} + p_{20} + p_{23},$$
$$p'_{12} = p_{01} + p_{12},\ p'_{13} = p_{01} + p_{13}.$$

So

$$N(B') = 8 \sum_{0 \le i < j \le 3} p'_{ij} = 8(-p_{00} + p_{01} + p_{12} + p_{13} + p_{23}).$$

Using lemma 1.11 (for $i = 0$), we get $N(B') = 8p_{01} + N(B)$. So,

$$N(B') - N(B) = 8p_{01} < 0,$$

which is a contradiction, since $N(B)$ is minimal. So, given some $p_{ij} < 0$, we are looking for a new superbasis B', such that $N(B') - N(B) = 8p_{ij} < 0$. Equivalently, we are looking for a new basis of $\mathcal{L}$, say V' such that $V' = U_3 V$, for some $U_3 \in SL_3(\mathbb{Z})$ having the additional property, each column of U_3 sums to zero (i.e. we want the basis to be extended to a superbasis of $\mathcal{L}$). In our case

$$U_3 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix}$$

We have five more cases.

- $p_{02} < 0$. We choose

$$U_3 = \begin{bmatrix} 0 & -1 & -1 \\ 0 & 1 & 0 \\ -1 & -1 & 0 \end{bmatrix}.$$

Then,

$$B' = [\mathbf{v}'_0, \mathbf{v}'_1, \mathbf{v}'_2, \mathbf{v}'_3]^T = [-\mathbf{v}_0, \mathbf{v}_0 + \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_0 + \mathbf{v}_3]^T,$$

and $N(B') - N(B) = 8p_{02} < 0$.

- $p_{03} < 0$. We choose

$$U_3 = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Then,

$$B' = [\mathbf{v}'_0, \mathbf{v}'_1, \mathbf{v}'_2, \mathbf{v}'_3]^T = [-\mathbf{v}_0, \mathbf{v}_0 + \mathbf{v}_1, \mathbf{v}_0 + \mathbf{v}_2, \mathbf{v}_3]^T,$$

and $N(B') - N(B) = 8p_{03} < 0$.

- $p_{12} < 0$.

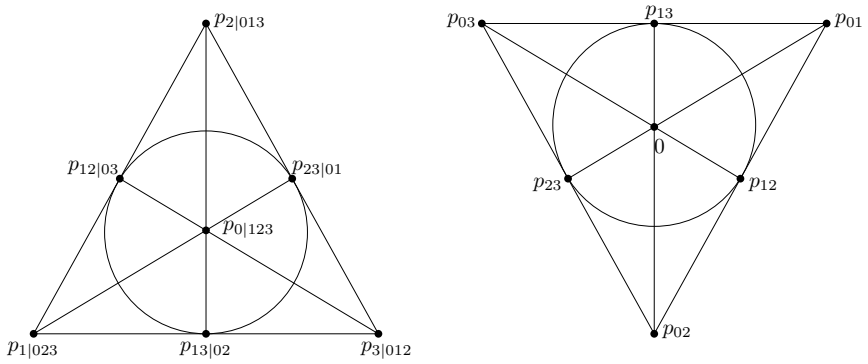


FIGURE 3. (a) The Fano plane annotated with the vonorms and (b) the dual Fano plane (Fano plane is self-dual) annotated with the conorms.

$p_{13} < 0$.
$p_{23} < 0$.
So we proved the following.

todo:check all the possible cases

Proposition 1.19. *A superbasis of a lattice $\mathcal{L} \subset \mathbb{R}^3$ that minimizes N is obtuse.*

Questions. (**i**). Is the inverse true? If we have an obtuse superbasis, does minimize N?1

Remark 1.20. If exactly one p_{ij}, $i \neq j$ is negative, then we can find an obtuse superbasis using the previous proof.

1.7. **An Algorithm for VFK for 3-dimensions.** We begin with the following definition.

Definition 9. A projective plane is a structure consisting of points and lines, where each line is a set containing at least two points, having the properties:
• two points are contained in a unique line
• two lines intersect in a unique point
• there exist four points with no three collinear.

In Figure 3 the structure of the Fano plane is shown.
We can see that the points and the lines have the following properties.
• there are seven points and seven lines
• each point lies on three lines, and each line contains three points
• each two points lie on a unique line, and each two lines intersect in a unique point.

Also, there is the dual projective space of the Fano plane. To construct the dual space we interchange the sets of points and lines, but preserve incidence. So the dual Fano plane is again a projective plane. In fact, it is again a Fano plane i.e., it is self dual.

1.7.1. *Labeling.* Say $\mathcal{L}$ be a full rank lattice of three dimension with basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. We call the seven subsums $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{123} = \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3$ putative Voronoi vectors,

their corresponding square Euclidean length, given by Formulas 1.3, are called putative vonorms and we denote them by $\mathcal{V}(\cdot)$. The corresponding conorms are called putative conorms for the putative superbase

$$\{\mathbf{v}_0 = -\mathbf{v}_1 - \mathbf{v}_2 - \mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\},$$

which will be correct if this superbase is indeed obtuse. In Figure 3(a) the (primal) Fano plane is shown annotated with the vonorms. In 3 dimensions there are 8 characters among which 1 improper (corresponding to the trivial map to 1) as well as 7 proper conorms. These are associated to the dual Fano plane as shown in Figure 3(b).

I think three vonorms lie at the same line when combining two gives the third. It is also related to the relation between cosets. Have to further look at that. The incidence relation between voronoi vectors is represented by the Fano plane.

In particular, the primal Fano plane contains $2^3 - 1 = 7$ vonorms. The points in the primal plane correspond to the vonorms

$$\mathcal{V}(\mathbf{v}_i) = \mathcal{V}(-\mathbf{v}_{jkl}) = \mathcal{V}(\mathbf{v}_{jkl}) = p_{i|jkl} = p_{jkl|i} = \mathbf{v}_i \cdot (\mathbf{v}_j + \mathbf{v}_k + \mathbf{v}_l)$$

and

$$\mathcal{V}(\mathbf{v}_{ij}) = \mathcal{V}(\mathbf{v}_{kl}) = p_{ij|kl} = p_{kl|ij} = (\mathbf{v}_i + \mathbf{v}_j) \cdot (\mathbf{v}_k + \mathbf{v}_l),$$

for distinct $i, j, k, l \in \{0, 1, 2, 3\}$. The points in the dual plane correspond to each line in the primal plane. These are the putative conorms and from Lemma 1.14, they are 0 and the six numbers p_{ij} for $0 \leq i < j \leq 3$.

The following property connects the primal and the dual plane in the case $\mathcal{L}$ is VFK, and allows the algorithm to work in the dual plane. In fact, Property 1.21 and Corollary 1.22 is what forces the relations, as represented by the lines between the points, between vonorms in the primal setting.

Is there some other intuition behind this association? Is this simply a geometric interpetation of the discrete Fourier trasform?

Property 1.21. *For all non-trivial characters in a VFK lattice, there exists a line in the primal plane on which all points have value* $+1$ *(*3 *in total) and the rest of the points (*4 *in total) have value* -1.

Proof. The proof is exhaustive, that is we must consider all cases based on Theorem 1.13 and Lemma 1.10. For example, let $S = \{\mathbf{v}_1, \mathbf{v}_2\}$. Then, $\chi(\mathbf{v}_0) = +1, \chi(\mathbf{v}_1) = -1, \chi(\mathbf{v}_2) = -1, \chi(\mathbf{v}_3) = +1, \chi(\mathbf{v}_{12}) = +1, \chi(\mathbf{v}_{13}) = -1, \chi(\mathbf{v}_{23}) = -1$. The $+1$s in this case correspond to the line going through points with vonorms $p_{12|03}, p_{0|123}, p_{3|012}$ as shown in Figure 3(a). □

Based on this property, one can prove the following corollary that connects the primal and the dual planes and it is the main tool that drives the design of the algorithm for discovering the Selling parameters that provide obtuse superbases. The idea is to work with the dual Fano plane, manipulating the conorms (Selling parameters) to get an obtuse superbasis.

Corollary 1.22. *For a VFK* $\mathcal{L}$ *lattice, the sum of the* 4 *numbers off a line* ℓ *on the primal plane minus the sum of the* 3 *numbers on the line* ℓ *is equal to* 4 *times the conorm of the point corresponding to line* ℓ *on the dual plane.*

Proof. By taking all cases, one can show that the statement is true. □

1.7.2. *The Algorithm.* The algorithm is given below. See appendix for a more detailed psuedocode.

Algorithm 1 : Sloane-Conway for rank 3
`INPUT: a basis` $\mathcal{B} = \{\mathbf{v}_i\}_{i=1,2,3} \subset \mathbb{Z}^3$
`OUTPUT: An obtuse superbasis` $\{\mathbf{v}'_i\}_{i=0,1,2,3}$

1: Construct the 7 lines $L = \{\ell_1, ..., \ell_7\}$ and the 7 points $\Phi = \{\phi_i\}$ of the dual Fano plane and associate to each point ϕ_i the respective putative conorm c_i
2: **While** there is a negative number c_i in some ϕ_i
3: Consider the line ℓ' containing ϕ_i and 0 and set $\varepsilon = |c_i|$
4: $\forall \phi_i \in \ell' : c_i \leftarrow c_i + \varepsilon$
5: $\forall \phi_i \notin \ell' : c_i \leftarrow c_i - \varepsilon$
6: Update Φ and the base $\mathcal{B}$
7: return $\mathcal{B}$

The putative conorms $\{c_i\}_i$ are given by the dot products $-\mathbf{v}_i \cdot \mathbf{v}_j$. If in the dual plane all conorms are non-negative, then the superbase is indeed obtuse. To achieve that, the algorithm moves between equivalent superbases for $\mathcal{L}$ correcting negative conorms in the dual Fano plane. If the superbasis is not obtuse then we shall always have a line which contains zero and a negative element. This is because we shall always have a zero to some line and this zero (a point on the dual Fano plane) is connecting with all other points. So it will connect also with the negative entry.

In line 1 of the algorithm, the dual Fano plane is set up. In particular, we compute all putative conorms by using the Formula 1.7. One of these putative conorms will be 0. Indeed, since the line in the primal plane going through points with vonorms $p_{12|03}, p_{23|01}, p_{13|02}$ corresponds to the center point in dual plane we get that its conorm will be

$$\tfrac{1}{4}\big(p_{3|012} + p_{1|023} + p_{2|013} + p_{0|123} - p_{12|03} - p_{23|01} - p_{13|02}\big) = 0.$$

Lines 3-5 are repeated as far as there is a negative conorm. As soon as all conorms become non-negative, then they are not putative any more and we can compute the vonorms and from there the obtuse superbase. To accomplish this, in Line 3 of the algorithm a line ℓ' is chosen such that it contains a negative conorm with absolute value ε as well as 0 - in case many such lines exist choose arbitrarily one of them. In Lines 4 and 5, ε is added to the conorms lying at the line ℓ' and is substracted from the other four conorms.

As soon as the Selling parameters for a lattice of the first kind have been identified. Let B be the superbase matrix, where each column is a base. The matrix B has n rows and $n+1$ columns. Then, we construct the inner product matrix for the superbase which is the following:

$$Q = B^T B = \begin{bmatrix} p_{0|12...n} & -p_{01} & -p_{02} & \cdots & -p_{0n} \\ -p_{10} & p_{1|02...n} & -p_{12} & \cdots & -p_{1n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ -p_{n0} & -p_{n1} & -p_{n2} & \cdots & p_{n|01...n-1} \end{bmatrix}$$

In this problem we know Q and we want to discover B. Notice that Q is diagonally dominant and positive definite. By using rank deficient Cholesky decomposition on Q we acquire B and as a result the obtuse base.

todo: Say sthm about this factorization. refs etc.

1.7.3. *Correcteness.* Let the putative obtuse base $B = \{\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ for the lattice $\mathcal{L}$ and let $B' = \{\mathbf{v}'_0, \mathbf{v}'_1, \mathbf{v}'_2, \mathbf{v}'_3\}$ be the base after an iteration of the algorithm. We will show that this is indeed a putative obtuse base for $\mathcal{L}$. Assume that there

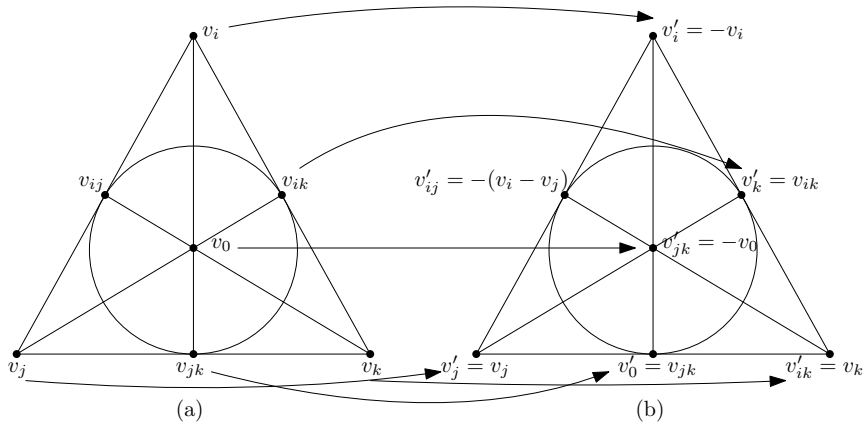


FIGURE 4. (a) The vonorms before the transformation (b) The vonorms after the transformation and their correspondence. Note that all of them apart one are equal up to sign.

exists $p_{ij} < 0$. This means that $\mathbf{v}_i - \mathbf{v}_j$ is shorter than $\mathbf{v}_{ij} = \mathbf{v}_i + \mathbf{v}_j$. Thus, the target of the algorithm should be to use the shorter putative Voronoi vector without changing considerably the other voronoi vectors. Assume that $i, j \neq 0$. Then, we choose the new base to be $\mathbf{v}'_0 = \mathbf{v}_0 + \mathbf{v}_i, \mathbf{v}'_i = -\mathbf{v}_i, \mathbf{v}'_j = \mathbf{v}_j$ and $\mathbf{v}'_k = \mathbf{v}_i + \mathbf{v}_k$ where $k \in \{0,1,2,3\} - \{0,i,j\}$, in order to use this new shorter vector as shown in Figure 4. If $i = 0$ (or j symmetrically) then we choose the new base B' to be $\mathbf{v}'_0 = \mathbf{v}_0, \mathbf{v}'_j = -\mathbf{v}_j, \mathbf{v}'_i = \mathbf{v}_i + \mathbf{v}_j$ and $\mathbf{v}'_k = \mathbf{v}_k + \mathbf{v}_j$, for $i, k \neq j$ in order to accomplish the same goal for the vector $\mathbf{v}_0 - \mathbf{v}_j$.

The following series of lemmas shows why the algorithm correctly handles the conorms alone in the manner described in the algorithm and moves among valid putative bases for $\mathcal{L}$ until the obtuse base is reached.

Lemma 1.23. *The new putative obtuse base B' for lattice $\mathcal{L}$ has 6 Voronoi vectors equal to B up to sign and only one different.*

Proof. Assume that $p_{ij} < 0$ and $i, j \neq 0$. The other case is similarly handled and we omit it. Figure 4 shows how these voronoi vectors are associated. Only one - the new shorter $\mathbf{v}'_{ij}$ - cannot be associated.

□

The 6 Voronoi vectors are equal up to sign and thus all of them maintain the same vonorms. This means that only one vonorm is changed in the vonorm space and all other remain the same. In addition, it is obvious that the new base B' generates the lattice $\mathcal{L}$ since it is a linear combination of the previous bases. The following lemma states that only one vonorm changes when moving from B to B'.

This is not enough, to get another Lattice basis.

Lemma 1.24. *Let $p_{ij} < 0$ and let $\varepsilon = |p_{ij}|$. The unique vonorm that changes when moving from base B to B' is decreased by 4ε.*

Proof. Let $i, j \neq 0$. We get:

$$\mathcal{V}(\mathbf{v}'_{ij}) = \mathcal{V}(\mathbf{v}_j - \mathbf{v}_i) = ||\mathbf{v}_j||^2 + ||\mathbf{v}_i||^2 - 2\mathbf{v}_i \cdot \mathbf{v}_j$$
$$\mathcal{V}(\mathbf{v}_{ij}) = \mathcal{V}(\mathbf{v}_j + \mathbf{v}_i) = ||\mathbf{v}_j||^2 + ||\mathbf{v}_i||^2 + 2\mathbf{v}_i \cdot \mathbf{v}_j$$

Then:

$$\mathcal{V}(\mathbf{v}_{ij}) - \mathcal{V}(\mathbf{v}'_{ij}) = 4\mathbf{v}_i \cdot \mathbf{v}_j = -4p_{ij} = 4\varepsilon$$

Note that all other vonorms remain the same after one iteration of the algorithm. The case where $i = 0$ is similarly handled. □

The following lemma, which proves the correcteness of the algorithm, states that we can move from superbase B to superbase B' by handling appropriately the conorms, that is working in the conorm (dual) space.

Lemma 1.25. *Let $p_{ij} < 0$ and let $\varepsilon = |p_{ij}|$. The conorm space of the new base B' is the conorm space of the previous base B in which we add ε to all conorms in the line going through p_{ij} and 0 and substract ε from all other conorms.*

Proof. This is a straightforward implication of Equation 1.7 that defines the conorm. This equation sums all vonorms on the line with positive sign and all vonorms off the line with negative sign. Thus, when the vonorm that has changed by 4ε according to Lemma 1.24 is on the line then the conorm of this line in the conorm space is reduced by ε (due to the $-\frac{1}{4}$ factor in Equation 1.7). When this vonorm is off the line then the corresponding conorm is reduced by ε. □

Lemma 1.25 proves the correctness of the iteration of the algorithm and shows why it works in the conorm (dual) space.

1.7.4. *Complexity Analysis.* For the analysis we assume integer lattices, that is, the coordinates of the vectors are all integers. The following property is a straightforward implication of how the new putative obtuse base B' is constructed from putative obtuse base B.

Property 1.26. *The putative obtuse base of the lattice $\mathcal{L}$ after each iteration has only integer coordinates.*

This property ensures that the unique putative Voronoi vector that is changed after each iteration is reduced by $4\varepsilon \in \mathbb{N}$ and $4\varepsilon \geq 1$. The following theorem states the time complexity of the algorithm.

Theorem 1.27. *Let $N = \max_{0 \leq i \leq 3}\{||\mathbf{v}_i||^2\}$. The time complexity of the algorithm that finds the obtuse superbase for an integer lattice in 3 dimensions is $O(N)$.*

Proof. In each iteration one vonorm is reduced by at least 1. There are in total $2^3 - 1 = 7$ vonorms and the maximum value for each such vonorm by definition is N, while the minimum value is 1. As a result, each vonorm can be reduced $O(N)$ times. Thus, the total number of iterations of the algorithm is $O(N)$. □

Note that this algorithm is pseudopolynomial since it is of exponential complexity w.r.t. the size of the input in bits, which is $O(\log N)$. Finally, before applying the previous algorithm we apply LLL or BKZ to reduce the size of the vector basis.

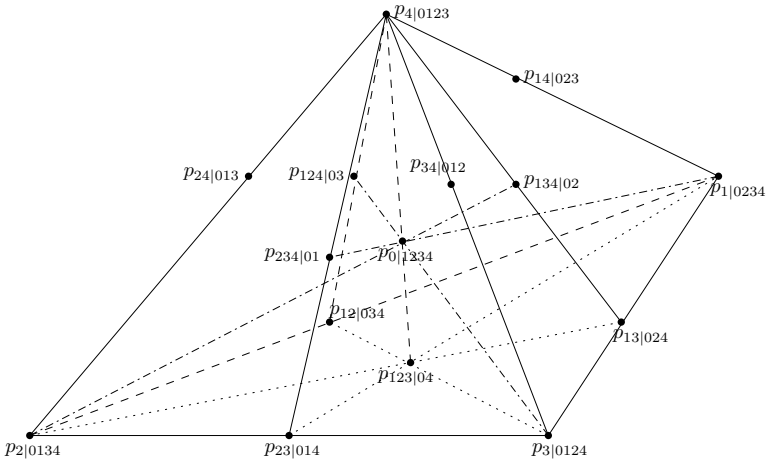


FIGURE 5. $PG(3,2)$ and the putative vonorms. Not all lines are depicted.

2. The 4-d Case

Let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ be the basis for a 4-d lattice Λ and its corresponding superbase $B = \{\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where $\mathbf{v}_0 = -(\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 + \mathbf{v}_4)$. We represent the incidence relations between the $2^4 - 1$ proper putative vonorms by using the 3-d projective space $PG(3,2)$. $PG(3,2)$ has 15 points, 35 lines and 15 planes. It has the following properties:

- A point is contained in 7 lines and 7 planes.
- Each line contains 3 points and is contained in 3 planes.
- Each plane contains 7 points and 7 lines.
- Each plane is isomorphic to a Fano plane.
- Every pair of distinct planes intersect at a point.
- A line and a plane not containing this line intersect at exactly one point.
- $PG(3,2)$ is a Desarguesian plane of finite order (2) and thus it is self-dual.

From Wikipedia under lemma "Duality (Projective Geometry)"

$PG(3,2)$ is depicted in Figure 5.

2.1. Labelling.

The labelling follows the same methodology with the 3-d case and is given in Figure 6 for each Fano plane.

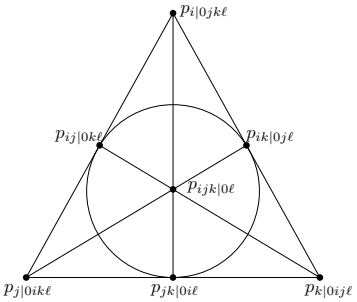
Similarly to Property 1.21 for 3-d we provide the following proposition.

Property 2.1. *For all non-trivial characters there exists a plane in the vonorm space on which all points have value* $+1$ *(*7 *in total) and the rest of the points (*8 *in total) have value* -1*.*

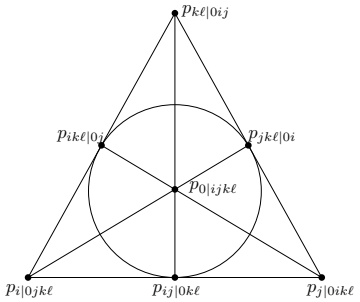
Proof. There are in total $2^4 - 1$ proper characters. The proof is similar to that of Property 1.21. □

Based on this property, we prove the following corollary that connects the primal and the dual projective spaces.

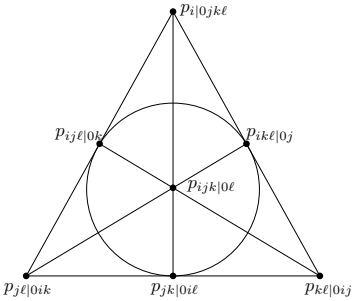
Corollary 2.2. *The sum of the* 8 *numbers off a plane* p *on the primal space minus the sum of the* 7 *numbers on the plane* p *is equal to* 8 *times the conorm of the point in the dual space that corresponds to plane* p*.*



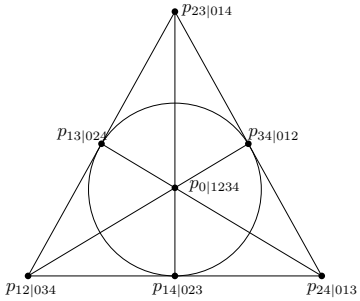
(a) 4 planes: i, j, k, ℓ is a permutation of $\{1, 2, 3, 4\}$



(b) 6 planes: i, j, k, ℓ is a permutation of $\{1, 2, 3, 4\}$



(c) 4 planes: i, j, k, ℓ is a permutation of $\{1, 2, 3, 4\}$



(d) 1 plane

FIGURE 6. We depict each Fano plane among the 15 possible and show how the putative vonorms label $PG(3, 2)$.

Proof. We take 4 cases based on the ones depicted in Figure 6.

Case (a): $(p_{\ell|0ijk} + p_{i\ell|0jk} + p_{j\ell|0ik} + p_{k\ell|0ij} + p_{ij\ell|0k} + p_{jk\ell|0i} + p_{ik\ell|0j} + p_{0|ijk\ell}) - (p_{i|0jk\ell} + p_{j|0ik\ell} + p_{k|0ij\ell} + p_{jk|0i\ell} + p_{ik|0j\ell} + p_{ij|0k\ell} + p_{ijk|0\ell}) = 8v_\ell(v_i + v_j + v_k + v_\ell) = 8(-v_\ell v_0) = 8p_{\ell 0}$

Thus, is this case we take 4 putative selling parameters that are of the form p_{x0}, where $x \in \{1, 2, 3, 4\}$.

Case (b): $(p_{k|0ij\ell} + p_{\ell|0ijk} + p_{ik|0j\ell} + p_{i\ell|0jk} + p_{j\ell|0ik} + p_{jk|0i\ell} + p_{ijk|0\ell} + p_{ij\ell|0k}) - (p_{i|0jk\ell} + p_{j|0ik\ell} + p_{ij|0k\ell} + p_{k\ell|0ij} + p_{jk\ell|0i} + p_{ik\ell|0j} + p_{0|ijk\ell}) = 8(-v_k v_\ell) = 8p_{k\ell}$

Thus, is this case we take 6 putative selling parameters that are of the form p_{xy}, where $x, y \in \{1, 2, 3, 4\}$, $x \neq y$ and $p_{xy} = p_{yx}$.

Case (c): $(p_{j|0ik\ell} + p_{k|0ij\ell} + p_{\ell|0ijk} + p_{ij|0k\ell} + p_{ik|0j\ell} + p_{i\ell|0jk} + p_{jk\ell|0i} + p_{0|ijk\ell}) - (p_{i|0jk\ell} + p_{ij\ell|0k} + p_{ijk|0\ell} + p_{ik\ell|0j} + p_{j\ell|0ik} + p_{jk|0i\ell} + p_{k\ell|0ij}) = 0$

Thus, is this case we take 4 zeros.

Case (d): $(p_{1|0234} + p_{2|0134} + p_{3|0124} + p_{4|0123} + p_{123|04} + p_{124|03} + p_{134|02} + p_{234|01}) - (p_{12|034} + p_{14|023} + p_{24|013} + p_{34|012} + p_{23|014} + p_{13|024} + p_{0|1234}) = 0$ □

If the superbase was obtuse then the putative conorms would be the correct Selling parameters, which would be 10 in total plus five zeroes as described in the preceding corollary. In Figure 7 the (dual) conorm space is shown with respect to the vonorm space of Figure 5.

QUESTION: How to move between adjacent superbases bases which are interpreted in the conorm space as a simple operation (like in 3d the $+-\varepsilon$) and have at the same time a nice monotonicity property? Is the

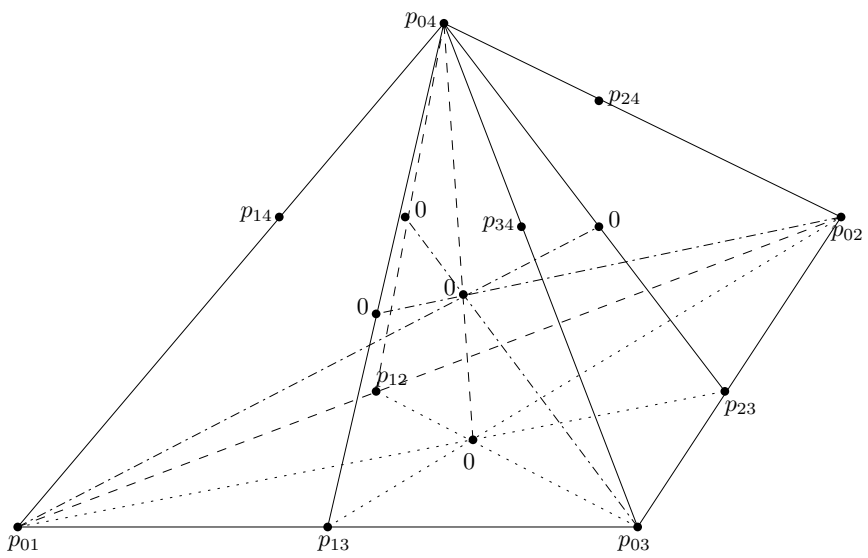


FIGURE 7. The conorm space for the vonorms of Figure 5. Not all lines are depicted.

existence of such a property what we are looking for that discerns the vfk from the non-vfk lattices?

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Appendix

Algorithm 2 : Pseudocode of Sloane-Conway algorithm for rank 3
INPUT: a basis $\mathcal{B} = \{\mathbf{v}_i\}_{i=1,2,3} \subset \mathbb{Z}^3$ of a lattice $\mathcal{L}$
OUTPUT: An obtuse superbasis $\{\mathbf{v}'_i\}_{i=0,1,2,3}$ of the lattice $\mathcal{L}$

$\mathbf{v}_0 \leftarrow -(\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3)$
$P = (p_{ij}) \leftarrow -\texttt{GramMatrix}(\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ # 4×4 symmetric matrix
$\Phi = \{p_{ij}\} \cup \{0\}$
the Lines of the dual Fano plane (see figure 3 (b))
$L_1 \leftarrow \{p_{03}, p_{13}, p_{01}\}$
$L_2 \leftarrow \{p_{03}, 0, p_{12}\}$
$L_3 \leftarrow \{p_{03}, p_{23}, p_{02}\}$
$L_4 \leftarrow \{p_{13}, 0, p_{02}\}$
$L_5 \leftarrow \{p_{01}, 0, p_{23}\}$
$L_6 \leftarrow \{p_{01}, p_{12}, p_{02}\}$
$L_7 \leftarrow \{p_{23}, p_{13}, p_{12}\}$ # the circle in Fano plane
While there is a negative element in L_i which contains also 0
  $\forall p \in L_i : p \leftarrow p + \varepsilon$
  $\forall p \in \Phi - L_i : p \leftarrow p - \varepsilon$
  Update the set of lines L_i
Update the matrix P
return The basis corresponding to P

We executed the previous algorithm with LLL reductions and without to the input basis. We summarized the results to the following table.

bits of N	without LLL	with LLL
100	3.9	2.1
200	3.7	2.08
300	4.15	1.88
400	3.98	2.26

Table 1. We executed 100 random instances for each row. We computed the average rounds for termination.